The Educational Knowledge Graph as the Cognitive Core of the AI-Native University

A scientific, architectural and pedagogical position paper on the DLU Educational Knowledge Graph (EKG)

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Introducing artificial intelligence as a layer on top of existing LMS/SIS platforms yields faster administration of an unchanged model. DLU adopts a different hypothesis: to place at the centre not content but knowledge represented as a graph, formally modelled and probabilistically traced. The DLU Educational Knowledge Graph (EKG) is the semantic backbone that makes curriculum, learning outcomes, concepts, competencies, assessments, evidence, mastery and career trajectories machine-understandable. This paper argues that the *EKG is the cognitive core* and *didactic intelligence of the system*, and that two modelling decisions — (i) representing prerequisites as a dependency graph (a DAG) and (ii) estimating learner knowledge as a probabilistic latent state with explicit uncertainty (mastery ≠ confidence) — are grounded in the knowledge-tracing, learning-analytics and learning-science literature. We formalise the learner model, present a concrete case (the knowledge graph of a Machine Learning course), and show numerically how knowledge about a student is constructed and how the aggregate learning state then powers faculty use cases.

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1. Motivation and scientific grounding

Bloom (1984) documented that mastery learning with one-to-one tutoring shifts outcome distributions by roughly two standard deviations relative to conventional classrooms — an effect historically unaffordable at scale, which AI now makes approachable [Bloom, 1984]. Operationalising mastery learning, however, requires three *measurable* ingredients that the classical university leaves implicit:

(a) a representation of the **structure of knowledge** (what depends on *what*),
(b) an **estimate of learner state** updated from evidence, and
(c) a **constructive alignment** of outcome–activity–evidence declared before content [Biggs, 1996].

Representing the structure of knowledge as a **graph with prerequisite relations** is the canonical form of an Educational Knowledge Graph: a prerequisite $A \rightarrow B$ means A should be mastered before B, and paths through the graph become learning routes and diagnostic instruments [Chen et al., 2018 (KnowEdu); Abu-Salih, 2021]. Encoding prerequisites as a **DAG** reflects Vygotsky's *zone of proximal development*: effective intervention acts on the concepts that are *ready* given the current state [Vygotsky, 1978].

Estimating a **latent knowledge state** has a long tradition: Bayesian Knowledge Tracing (BKT) models per-skill mastery as a two-state hidden Markov model with four parameters — prior, learn, guess and slip [Corbett & Anderson, 1995]; its logistic extensions (PFA, IRT, Elo) and neural variants (Deep Knowledge Tracing) extend power and flexibility [Pelánek, 2017; Piech et al., 2015; Abdelrahman et al., 2023]. DLU adopts a **Bayesian Beta** posterior — simpler to explain than BKT, but, unlike classical BKT, it **exposes the uncertainty** of the estimate (the variance of the Beta), realising an inspectable and contestable *open learner model* [Bull & Kay, 2010].

Finally, the **physics of memory** — Ebbinghaus's forgetting curve, the testing effect, and desirable difficulties — motivates temporal decay and the choice of effortful activities: spaced repetition and active retrieval improve long-term retention [Ebbinghaus, 1885; Roediger & Karpicke, 2006; Bjork & Bjork, 2011; Settles & Meeder, 2016].

The design consequence is sharp: **progression is evidence-driven, not time-based**, and the unit of truth is the latent state (mastery,confidence) per concept, propagated along the graph toward module/course outcomes (MLO/CLO) and competencies.

2. The formal learner model

2.1 Evidence and its weighting

Every activity produces **evidence** with a normalised score $s \in [0,1]$ and an effective weight combining instrument quality, evaluator reliability, recency and authenticity:

$$w_e = \text{quality} \cdot \text{reliability} \cdot \text{recency} \cdot \text{authenticity}$$

$$\text{effectiveWeight} = \text{baseWeight} \cdot \text{reliability} \cdot \text{authenticity} \cdot \text{recency} \cdot \text{difficultyCalibration}$$

DLU distinguishes three classes:

- **QualifiedEvidence** (graded item, rubric, authenticated assessment) updates high-stakes mastery;
- **TutorEvidence** (a tutor micro-assessment) carries a *capped* weight;
- **InformalSignal** (hesitation, explanation quality) drives only diagnosis and planning. *TutorEvidence cannot override authoritative grades.*

2.2 Mastery estimation (Beta posterior)

For a pair (learner u, target k) we maintain a Beta (α, β). Given evidence (s, w_e):

$$\alpha \leftarrow \alpha + w_e\, s, \qquad \beta \leftarrow \beta + w_e\,(1-s)$$

$$M = \frac{\alpha}{\alpha+\beta}, \qquad \mathrm{Var} = \frac{\alpha\beta}{(\alpha+\beta)^2\,(\alpha+\beta+1)}, \qquad C = 1 - 2\sqrt{\mathrm{Var}}$$

M (mastery) and C (confidence) **must never be collapsed**: $(M = .90, C = .25)$ ("promising but under-evidenced") differs from $(M = .90, C = .95)$ ("stable strength") [Bull & Kay, 2010]. Temporal decay is applied at query time to immutable observations (analogous to the forgetting curve / half-life regression) [Settles & Meeder, 2016].

Relation to BKT. BKT estimates $P(L)$ via a hidden Markov model with guess/slip [Corbett & Anderson, 1995]. The Beta formulation is a conjugate relative in which slip/guess are absorbed into $w_e\, s$; the added value is the **explicit variance** as a confidence measure — absent from single-valued BKT and crucial for deciding *when more evidence is needed before recommending*.

2.3 Hierarchical propagation and prerequisite readiness

Concept mastery propagates toward MLO → CLO → Skill with weights π and is **capped** by prerequisite readiness (DAG):

$$\mathrm{readiness}(k) = \prod_{p\,\in\,\mathrm{pre}(k)} E\,(p), \qquad M(\mathrm{MLO}_j) = \Big(\sum_k \pi_{jk}\; M(k)\Big) \cdot \min(1,\ \mathrm{readiness}(\mathrm{MLO}_j))$$

where E is the **confidence-sensitive effective mastery**:

$$E = M(\gamma + (1-\gamma)\,C), \quad \gamma \in [0,1], \qquad \mathrm{gap}(\theta) = \max(0,\ \theta - E)$$

2.4 Selecting the didactic action (Adaptive Tutor)

The Pedagogical Policy Engine (ATA) selects the **Next Best Learning Action (NBLA)** by maximising a versioned score; the LLM renders, it does not decide:

$$\mathrm{Score}(a) = \sum_{i=1}^{6} w_i\ \phi_i(a)\ - \sum_{i=7}^{9} w_i\ \psi_i(a)$$

with rewards $\phi =$ {ExpectedLearningGain, GoalAlignment, PrerequisiteReadiness, GapPriority, RetentionValue, EngagementPro
and penalties $\psi =$ {CognitiveLoad, Redundancy, RiskPenalty}. The objective is **learning gain and retention**, not engagement; policy changes require offline evaluation and controlled roll-out.

2.5 Path optimisation (Academic GPS)

The deterministic path orders items by marginal utility:

$$\mathrm{utility}(item) = \mathrm{gapCoverage} \cdot \mathrm{expectedGain} \cdot \mathrm{confidenceNeed} \cdot \mathrm{relevance} - \mathrm{timePenalty} - \mathrm{redundancyPenalty}$$

LLMs narrate the path; they do not rank it — a deliberate separation between deterministic ranking and generation.

3. A concrete case: the knowledge graph of a Machine Learning course

Consider **AI-501 — Foundations of Machine Learning** (EQF 7). Figure 1 shows an excerpt of the concept sub-graph (`Concept` nodes) with `PREREQUISITE_OF` relations (a DAG), the anchoring to outcomes (`MLO`/`CLO`) and to assessments (`Assessment`). Concept and prerequisite extraction can be automated with established techniques (concept extraction plus prerequisite mining) [Chen et al., 2018; ACE, 2024].

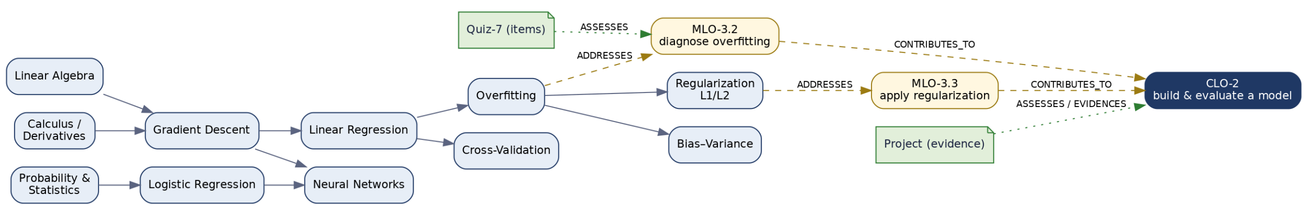

Figure 1 — Excerpt of the AI-501 Educational Knowledge Graph: concept prerequisite DAG (blue), module/course outcomes (gold/navy) and assessments (green). Edges: PREREQUISITE_OF, ADDRESSES, CONTRIBUTES_TO, ASSESSES/EVIDENCES.

Underlying EKG semantics (excerpt): `Course AI-501 —HAS_MODULE→ Module-3 —HAS_LESSON→ Lesson-3.2 —TEACHES→ Concept:Overfitting`; `Lesson-3.2 —ADDRESSES→ MLO-3.2 —CONTRIBUTES_TO→ CLO-2`; `Quiz-7.item —ASSESSES→ MLO-3.2`; `Evidence(project) —EVIDENCES→ CLO-2`; `Concept:LinearRegression —PREREQUISITE_OF→ Concept:Overfitting`. The **acyclicity** of prerequisites is a graph invariant checked in CI.

4. Use case A — How knowledge about a student is constructed (worked example)

Student u on the concept **Overfitting**. Non-informative prior $\alpha_0 = \beta_0 = 1$ (hence $M = 0.5$, high uncertainty). Three pieces of QualifiedEvidence are observed during Module 3, with $\Delta\alpha = w_e s$ and $\Delta\beta = w_e(1 - s)$:

1. MCQ, correct — $s = 1.0$, $w_e = 0.8 \Rightarrow \Delta\alpha = 0.80$.
2. MCQ, incorrect — $s = 0.0$, $w_e = 0.7 \Rightarrow \Delta\beta = 0.70$.
3. Rubric-scored open answer, good — $s = 0.9$, $w_e = 1.0 \Rightarrow \Delta\alpha = 0.90,\ \Delta\beta = 0.10$.

Posterior: $\alpha = 1 + 0.80 + 0.90 = 2.70$, $\beta = 1 + 0.70 + 0.10 = 1.80$.

$$M = 2.70/4.50 = 0.600, \qquad \mathrm{Var} = 2.70 \cdot 1.80/4.50^2 \cdot 5.50 = 4.86/111.375 = 0.0436$$

$$\mathrm{SD} = \sqrt{0.0436} = 0.209, \qquad C = 1 - 2(0.209) = 0.582$$

Effective mastery ($\gamma = 0.5$): $E = 0.600\,(0.5 + 0.5 \cdot 0.582) = 0.600 \cdot 0.791 = 0.474$. Against target $\theta = 0.75$: gap $= 0.276$.

Prerequisite effect. If upstream readiness is high — $E(\text{LinearRegression}) = 0.82$, $E(\text{GradientDescent}) = 0.88$ — propagation toward **MLO-3.2 "diagnose overfitting"** is not capped: readiness(MLO-3.2) $= \min(1,\ 0.82 \cdot 0.88) = 0.72$; with $M(\text{Overfitting}) = 0.60$, $M(\text{BiasVariance}) = 0.55$ and weights $\pi = [0.6, 0.4]$: $M(\text{MLO-3.2}) = (0.6 \cdot 0.60 + 0.4 \cdot 0.55) \cdot 0.72 = 0.58 \cdot 0.72 = 0.418$. Had Linear Regression been weak ($E = 0.40$), the cap would drop (readiness ≈ 0.35) and the system would **attribute the weakness to the prerequisite**, not to the current concept — the basis of faculty diagnosis (§5).

Tutor action. With a medium-high gap on Overfitting and *ready* prerequisites, the policy favours the sequence **WORKED_EXAMPLE → PRACTICE → REASSESS**; with weak prerequisites it would favour **REVIEW_PREREQUISITE** (the $w_3 \cdot \text{PrerequisiteReadiness}$ term dominating). The choice is **explainable** (goal + state + graph dependencies) and grounded in resources via GraphRAG [Edge et al., 2024], with learning gain as the objective [Roediger & Karpicke, 2006].

Each step appends immutable observations; the `mastery.updated` event rebuilds the student read projection. The student **sees and can contest** their own (M, C) state per concept — an open learner model [Bull & Kay, 2010].

5. Use case B — How the aggregate learning state serves faculty

Faculty do not work turn-by-turn but on the **aggregate** of the cohort's learning states, projected into a *Faculty Lens* (never the whole graph).

5.1 Outcome health

For an outcome o and a cohort of N students we define attainment and mean confidence:

$$\text{Attainment}(o) = \frac{1}{N}\sum_{u} \mathbf{1}\,[E_u(o) \geq \theta(o)], \qquad \overline{C}(o) = \frac{1}{N}\sum_{u} C_u\,(o)$$

An outcome with **low attainment and high mean confidence** is a *confirmed gap* (priority intervention); **ambiguous attainment with low confidence** signals *under-evidencing* — more assessment is needed, not re-teaching. This is the operational translation of "mastery ≠ confidence" at the cohort level.

5.2 Root-cause analysis on the prerequisite graph

Selecting a weak CLO (e.g. CLO-2), the system **decomposes** along `CONTRIBUTES_TO` to the MLOs, then to concepts, and computes for each prerequisite concept p an **attribution** score (where knowledge transfer breaks down):

$$\text{Blame}(p) = \text{blockedFraction}(p) \cdot \text{prereqCentrality}(p) \cdot (\theta - \overline{E}(p))_+$$

$$\text{blockedFraction}(p) = \frac{1}{N}\sum_{u} \mathbf{1}\,[\text{readiness}_u(\text{child}(p)) \text{ capped by } E_u(p) < \tau]$$

The instructor receives an ordered explanation — *contributing MLOs → concepts → highest-Blame prerequisites → assessment coverage → evidence confidence* — instead of a generic graph explorer.

Example: if 60 % of students with a weak CLO-2 have $E(\text{LinearRegression}) < 0.5$, the highest **Blame** falls on *Linear Regression*: efficient remediation is upstream, not on "overfitting".

5.3 Assessment validity and coverage

The EKG checks constructive-alignment invariants [Biggs, 1996]:

$$\text{Coverage} = \frac{|\{\, o : \exists \text{ Assessment ASSESSES } o \,\}|}{|\text{published outcomes}|}$$

and, at item level, **discrimination** flags weak questions via the point-biserial correlation between item outcome and total score:

$$r_{pb} = \frac{M_1 - M_0}{s_x}\sqrt{p\,(1-p)}$$

with p the proportion of correct responses. Items with low or negative r_{pb} warrant revision; this dialogues with psychometric theory (IRT 2PL: item *difficulty* b and *discrimination* a) used to calibrate items [Lord, 1980; van der Linden & Hambleton, 1997]. The result: the instructor **controls knowledge transfer** across four levers — outcome health, prerequisite root-cause, assessment validity/coverage, and content coverage — while retaining pedagogical authority (agents operate within inspectable envelopes).

5.4 From faculty to programme and institution

The same structures scale up: **Course × Outcome** / **Course × Competency** matrices, prerequisite integrity and orphan concepts for the dean; the **Programme → CLO → MLO → Lesson → Assessment → Evidence** traceability as a *view* for the provost, feeding accreditation indicators (in Italy, AVA / Ministerial Decree 1154) — continuous rather than annual controls.

6. Why these choices, in scientific summary

DLU choice	Grounding	References
Prerequisites as a queryable **DAG**	prerequisite structure of EKGs; ZPD	Chen 2018; Vygotsky 1978
Latent state (M, C) **with uncertainty**	knowledge tracing + open learner model	Corbett & Anderson 1995; Pelánek 2017; Bull & Kay 2010
Evidence-driven progression (mastery)	2-sigma / mastery learning	Bloom 1984
Temporal **decay**, spaced practice	forgetting curve, HLR, testing effect	Ebbinghaus 1885; Settles & Meeder 2016; Roediger & Karpicke 2006
Effortful, not frictionless activity	desirable difficulties	Bjork & Bjork 2011
Constructive alignment outcome→activity→evidence	instructional alignment	Biggs 1996
Graph-grounded tutoring	RAG over knowledge	Edge 2024; VanLehn 2011

(GraphRAG); explicit pedagogy	graphs; tutoring efficacy	
Neural (DKT) as an *estimator*, not pedagogical truth	DKT and interpretability limits	Piech 2015; Abdelrahman 2023

Epistemic caveat. DKT achieves strong predictive power but is less interpretable; DLU admits it as an **estimation source** inside the governed Bayesian engine, never as the owner of pedagogy.

7. Conclusion

Representing knowledge as a graph and the student as an uncertain latent state is not an engineering flourish: it is the condition for operationalising, at scale, established theories of learning. For the student the EKG is mirror and guide (open learner model plus adaptive tutoring); for the instructor it is the instrument that makes **knowledge transfer observable and governable** (outcome health, prerequisite root-cause, assessment validity); for the programme and the institution it is assurance of integrity and proof of attainment. It is, properly, the **didactic intelligence** of the AI-Native University.

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Formulae are illustrative and calibrated on DLU data; weights (w_i, π, γ) and thresholds are versioned, governed parameters (rule packs), not hard-coded constants.